

L02: Image Processing Fundamentals

Section 1: Technical Understanding

Among the most interesting lessons learned from this lab is that computer systems do not interpret a digital image as a continuous entity, but rather as an array with well-defined values and dimensions. An ordinary colored image, I found out, is a three dimensional array of Height $\times$ Width $\times$ Channels in which each pixel is assigned an 8-bit value of intensity levels from 0 to 255. Seeing the raw arrays of pixels compared to the more humanly readable images helped me see how easily numbers corresponded to shapes, colors and even shadows.

Learning about point operations and neighborhood operations revealed a bit more about how all this works: brightness and contrast manipulation are based on arithmetic, while spatial filtering requires use of convolution kernels analyzing neighboring pixel areas in order to calculate new values. In particular, creating custom convolution kernels used for edge detection and blurring was the most mentally challenging part of this process. Understanding how a 3×3 area of weights is moving across an image array in order to amplify or filter certain frequencies took some thought.

Section 2: Connections and Applications

The experiments performed during this lab were based on the basic concepts illustrated by the Nano Banana demo, showing how important pixel level manipulations are in setting the base for perception at the high level. In much the same way as classical filters perform extraction of edges, contrast, and colors, current AI-based models use the same principles in order to make

sense of the images. In practice, these techniques prove to be important for automated computer vision systems used for various tasks, including advanced medical image analysis, object detection in autonomous driving, camera enhancements in mobile applications, and facial recognition technology. As far as the future lab work is concerned, it seems that the combination of classical algorithms and models proves to be highly effective. Preprocessing steps like histogram equalization and filtering reduce noise in the input and prepare matrices for learning.

Section 3: Personal Reflection

With regard to the future, the field of image processing that most intrigues me is feature extraction and how deep learning networks automatically learn how to apply optimal filters, as opposed to manually creating kernels such as we did in this lab. This lab opened my eyes to how every filter on Instagram, every HDR image from a phone, and every auto-focusing feature has to do with a series of complex matrix manipulations done instantly behind the scenes of digital photography. Nevertheless, this discovery does not solve all of my curiosity, and I still wonder how current neural networks deal with highly corrupted images or occluded areas while still being able to recognize features.